

Computer Science - Answer Script

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Q1. Define a variable in programming and give an example.

A variable is a named storage location in memory used to hold a value that can change during program execution. Example: In Python, `age = 21` creates a variable called `age` that stores the value 21. The variable can later be reassigned, for example `age = 22`.

Q2. Differentiate between a compiler and an interpreter.

A compiler translates the entire source code into machine code before execution, producing an executable file, and reports all errors after scanning the whole program. An interpreter, on the other hand, translates and executes code line by line, stopping as soon as it hits an error. Compiled programs generally run faster, while interpreted programs are easier to debug and test.

Q3. What is the difference between a list and a tuple in Python?

A list is a mutable ordered collection, meaning its elements can be changed, added, or removed after creation, and is defined using square brackets, e.g. `[1, 2, 3]`. A tuple is an immutable ordered collection defined using parentheses, e.g. `(1, 2, 3)`, and once created its elements cannot be modified. Because tuples are immutable, they are generally faster and are often used for fixed data such as coordinates.

Q4. Explain the concept of a loop with an example.

A loop is a control structure that repeats a block of code multiple times until a specified condition is met. For example, a for loop can be used to print numbers from 1 to 5:

```
for i in range(1, 6):  
    print(i)
```

This executes the print statement five times, once for each value of `i`.

Q5. What is an algorithm? List its key characteristics.

An algorithm is a finite, well-defined sequence of steps used to solve a problem or perform a computation. Key characteristics include: it must be finite (terminate after a limited number of steps), definite (each step must be precisely defined), have zero or more well-specified inputs, produce one or more outputs, and be effective (each step must be basic enough to be carried out).